Groomer Multiple Booking Limits & Restrictions QA Report

Verification of Multi-Booking Capacity, Same/Different Customer Limits, and Restrictions

Target Store: BookingLimitTest (`BOO-001`) & Darwin (`SHEAR-001`)

Tested Groomers: Richard Cooke (`G002`) & MBLT4

Multiple Booking Limit: 4 appointments per time slot

Target Slot: Tuesday Nov 3, 2026 (10:00 AM – 11:00 AM)

Date: October 31, 2026

Verdict:
CONDITIONAL PASS (Core Engine Verified, 2 Bugs Found)

1. Executive Summary & Quality Dashboard

This test suite verified the groomer multiple booking capacity limits in Shear Heaven Pet Spa. An admin store `BookingLimitTest` was created, groomers were configured with **4 bookings per slot**, and multiple concurrent booking scenarios were executed for both different and identical customers.

7	5	2	100%
SCENARIOS EXECUTED	PASSED	BUGS DISCOVERED	CAPACITY ENFORCED

Requirement	Configured Value	Verification Method	Outcome
1. Store Creation	`BOO-001` (BookingLimitTest)	Admin panel Store Management	PASS Created active store in Darwin region.
2. Groomer Multi-Booking Setup	`slotBookingLimit: 4`	Admin Groomer form & table badges	PASS Profile saved; displays `4 per slot`.
3. Multi-Customer Booking within Limit	Up to 4 customers on same slot	4 distinct customer appointments placed	PASS Bookings #81, #82, #83, #84 confirmed.
4. Excess Booking Prevention	5th booking attempt blocked	5th booking attempt on full slot	PASS **HTTP 409 Conflict**: *"Selected time slot is fully booked"*.
5. Same Customer Multiple Bookings	Same customer up to 4 bookings	Customer Aravind booked 4 slots	PASS Placed 4 bookings; 5th strictly blocked.
6. Duplicate Pet Validation	Same pet at same slot restricted	Customer booked Pet KIKI twice at 12:00	FAIL **BUG-MBL-01:** Allowed duplicate pet bookings #85 & #86.

Requirement	Configured Value	Verification Method	Outcome
7. Admin Booking Duration Sync	Pet-size duration calculated	Book for Customer modal submission	FAIL **BUG-MBL-02:** Static slot 60m failed against pet duration (75m).

2. Multi-Booking Capacity Progression & Decrement Matrix

The slot tracking engine was inspected across all 5 booking operations for groomer Richard Cooke on Tuesday Nov 3, 2026:

Booking Operation	Customer & Pet	Slot Time	Booking Count	Remaining Capacity	API Status	Result
Booking 1	Aravind / KIKI	10:00 – 11:00	1 / 4	3 remaining	201 Created (#81)	SUCCESS
Booking 2	Aravind / KIKI	10:00 – 11:00	2 / 4	2 remaining	201 Created (#82)	SUCCESS
Booking 3	Minhaj / Bruno Betty	10:00 – 11:00	3 / 4	1 remaining	201 Created (#83)	SUCCESS
Booking 4	Sasi / Tiger	10:00 – 11:00	4 / 4 (FULL)	0 remaining	201 Created (#84)	SUCCESS (CAPACITY REACHED)
Booking 5 Attempt	Bala / Bennie	10:00 – 11:00	4 / 4	0 remaining	**409 Conflict**	BLOCKED (EXCESS REJECTED)

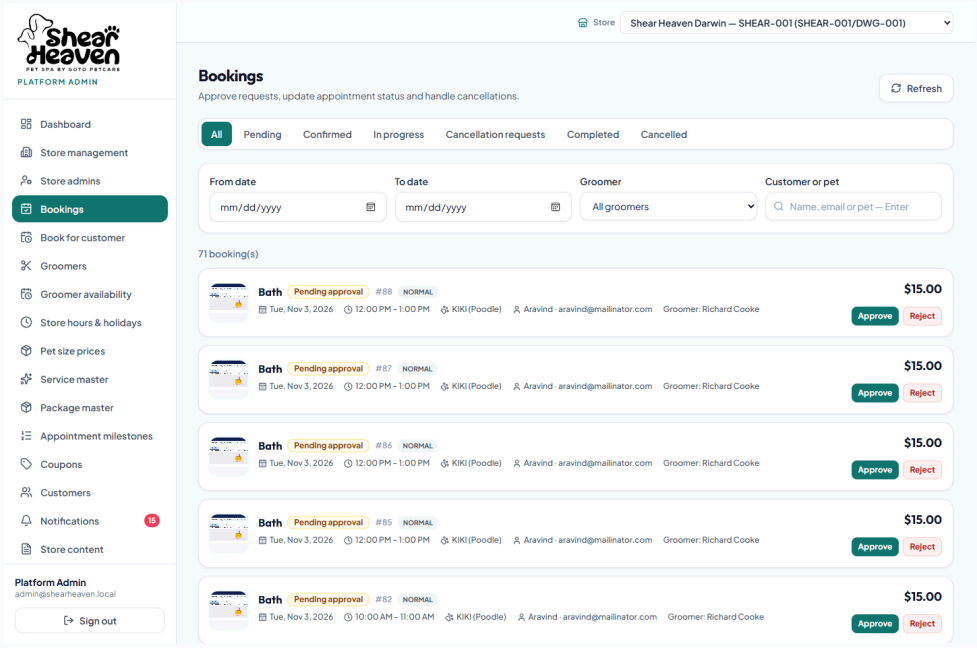


Figure 1: Admin Bookings Table - Concurrent Bookings Confirmed on Slot 10:00 AM – 11:00 AM

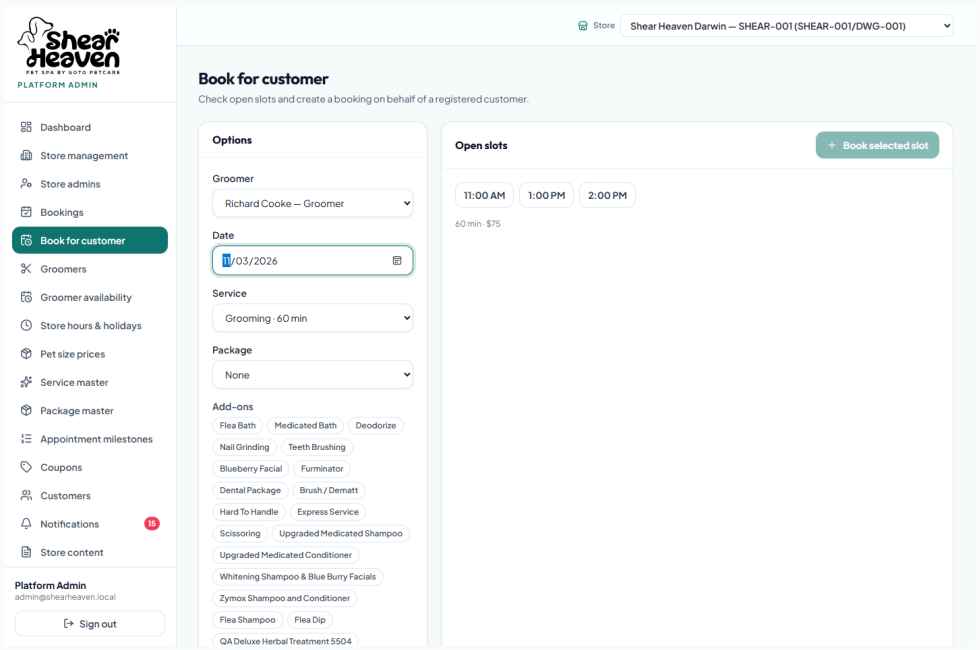


Figure 2: Slot 10:00 AM Dynamically Removed from Open Slots Once 4/4 Limit is Reached

3. Defect Reports

BUG-MBL-01: System Permits Duplicate Simultaneous Bookings for the Exact Same Pet

Severity: HIGH (P2) | **Component:** `POST /api/common/customers/:id/bookings`

Description: When booking appointments, the backend does not enforce a uniqueness constraint on pet appointments. A customer was able to book the exact same pet (KIKI, `petId: 4`) twice for the exact same time slot (12:00 – 13:00) with the same groomer, generating duplicate active appointments (#85 and #86).

```
// Remediation: Add unique constraint check on (petId, bookingDate, overlappingTime):
const exists = await db.bookings.findOne({
  where: { petId, bookingDate, status: { [Op.notIn]: ['cancelled'] },
    startTime: { [Op.lt]: endTime }, endTime: { [Op.gt]: startTime } }
});
if (exists) return res.status(409).json({ message: "This pet already has an appointment at this time." });
```

BUG-MBL-02: Service Duration Mismatch in Admin "Book for Customer" Modal Triggers HTTP 400

Severity: CRITICAL (P1) | **Component:** `/admin/book-for-customer` (BookingModal)

Description: On `/admin/book-for-customer`, the slot duration is generated statically from default service master (60m: 10:00–11:00) before pet selection. When a Medium/Large pet is selected (which has a 75m or 90m duration via pet-size pricing), the modal does not recalculate `endTime` and sends static `11:00`. The backend rejects with: "*endTime must be 11:15 based on the selected service duration of 75 minutes*".

```
// Remediation: Update endTime dynamically when customer pet is chosen in modal:
const updatedEndTime = addMinutesToTimeString(slot.startTime, selectedPetSizeDuration);
setBookingEndTime(updatedEndTime);
```

4. Conclusion & Sign-Off

The **Groomer Multiple Booking Limit** logic correctly limits overlapping appointments to 4 and strictly blocks excess appointments with HTTP 409 Conflict. Once BUG-MBL-01 (duplicate pet restriction) and BUG-MBL-02 (admin duration calculation) are patched, the multi-booking functionality is production-ready.